1(a)	work done per unit mass	**B1**
	work done moving mass from infinity (to the point)	**B1**
1(b)(i)	gravitational force provides centripetal force	**C1**
	$mv^2 / r = GMm / r^2$ and $v = 2\pi r / T$ OR $mr\omega^2 = GMm / r^2$ and $\omega = 2\pi / T$ OR $r^3 = GMT^2 / 4\pi^2$	**C1**
	$r^3 = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times (13.7 \times 24 \times 3600)^2 / 4\pi^2$ so r = 2.4×10^8 m	**A1**
1(b)(ii)	(E_P = –) GMm / r work done = $GMm / r_1 - GMm / r_2$	**C1**
	$= 6.67 \times 10^{-11} \times 360 \times 6.0 \times 10^{24} (1/6.4 \times 10^6 - 1 / 2.4 \times 10^8)$	**C1**
	= $\mathbf{2.2 \times 10^{10}}$ J	**A1**
1(b)(iii)	$g = GM / r^2$	**C1**
	ratio $= r_{TESS}^2 / r_{earth}^2$ $= (2.4 \times 10^8 / 6.4 \times 10^6)^2$ = **1400**	**A1**